Discrete Bounded Envy-Free Cake Cutting Protocol for 4 agents

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Outline

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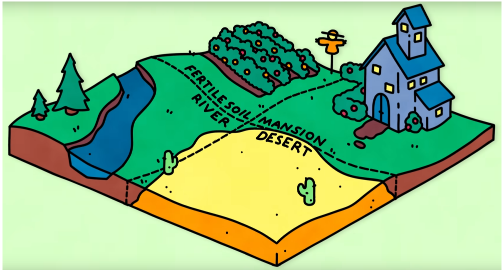
What is Cake Cutting?

- Cake is a metaphor for some heterogeneous resource that has to be divided among n people



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- Active research area in fair division and computer science

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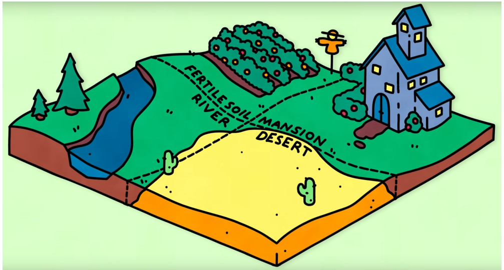

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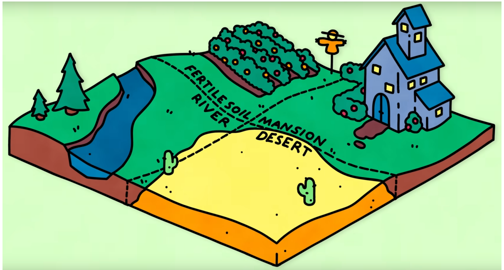

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Cake Cutting Model

- The cake C is an infinitely divisible resource represented by the interval $[0, 1]$
- A piece of cake is union of subintervals of $[0, 1]$
- Each agent i in $N = 1, \ldots, n$ has his own value function $V_i : [a, b] \subseteq [0, 1] \to \mathbb{R}^+$
- The value function V_i satisfies the following properties
  1. $V_i(\emptyset) = 0$
  2. $V_i([0, 1]) = 1$ **(Normalization)**
  3. $V_i(A \cup B) = V_i(A) + V_i(B) \quad \text{if } A \cap B = \emptyset$ **(Additivity)**

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What it means to be fair?

Defining Fairness

Let $X_i = \bigcup_j [a_j, b_j]$ be the final allocation to agent i

- **Proportional division**:
  - The first fairness criterion conceived by Stefan Banach and Bronisław Knaster in 1944
  - Each agent i gets at least $\frac{1}{n}^{th}$ of the total value in their valuation i.e. $V_i(X_i) \geq \dfrac{V_i(C)}{n}$
- **Envy-Free division**:
  - A stronger notion of fairness introduced by George Ganow and Marvin Stern in 1958
  - Each person should receive a share that in their eyes is at least as good as the share received by others i.e. $V_i(X_i) \geq V_i(X_j) \forall j \in N/\{i\}$

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Do such allocation exist?

Hugo Steinhaus, along with Banach and Knaster, proved that "an envy-free allocation always exists for n agents" provided the value functions satisfy the properties mentioned before.

The Cake Cutting Problem

Given a cake, an infinitely indivisible heterogeneous resource C, and a set of n agents N with each agent's value function satisfying the mentioned properties. Find an **envy-free** protocol to share the cake fairly between the n agents.

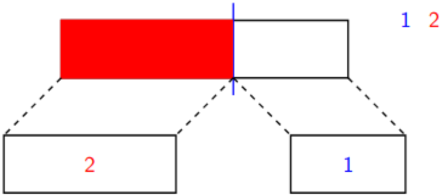
Outline

Cut-and-Choose

Even mentioned in the Bible, in the Book of Genesis(chapter 13), where Abraham divides the land of Canaan and Lot chooses first.

The Cut-Choose algorithm

- One agent(say P_1) cuts the cake into equal pieces according to his value function
- The other agent(P_2) chooses the piece he wants and the remaining piece is given to P_1



Why is it envy-free?

- P_1 is not envious of P_2 because he got to choose first, and he will choose the piece with the largest valuation according to his value function
- P_2 is not envious of P_1 because he values both pieces as the same

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Few Important Takeaways

- The cutter is always happy to choose last provided he gets a complete piece
- The agent who chooses first is always not envious of other agents as he has an irrevocable advantage over them

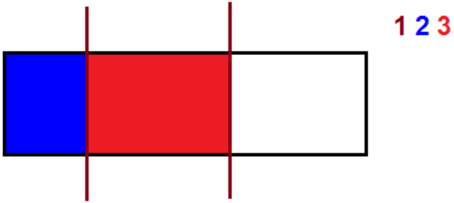
Outline

Protocol for 3 agents

Let the players be P_1, P_2, and P_3. The procedure is as follows:

Selfridge-Conway Protocol

- P_1 cuts the cake into 3 pieces of equal value according to V_1.
- If P_2 and P_3's favorite pieces are different, each player receives their favorite piece, and the process ends.



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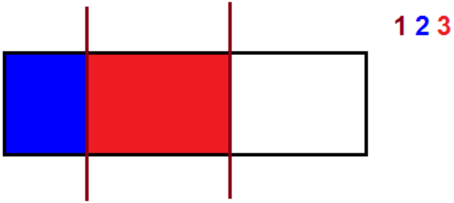

Protocol for 3 agents

Selfridge-Conway Protocol Continued

- Else, P_2 ranks the pieces in order of preference as A, B, C.
- P_2 cuts A into A_1 and A_2 such that $V_2(A_1) = V_2(B)$.
- P_3 chooses their favorite piece from $\{A_1, B, C\}$.
  - If P_3 does not choose A_1, P_2 is forced to take A_1.
- Now, the remaining task is to split A_2:
  - P_1 "dominates" the player who chose A_1 (denoted P_A). P_1 will never envy P_A regardless of how A_2 is allocated.
  - To ensure P_1 does not envy P_B, P_B divides A_2 into 3 pieces of equal value and P_1 is given higher priority in choosing a piece.
- The players, in the order P_A, P_1, P_B, pick their most preferred pieces of A_2.

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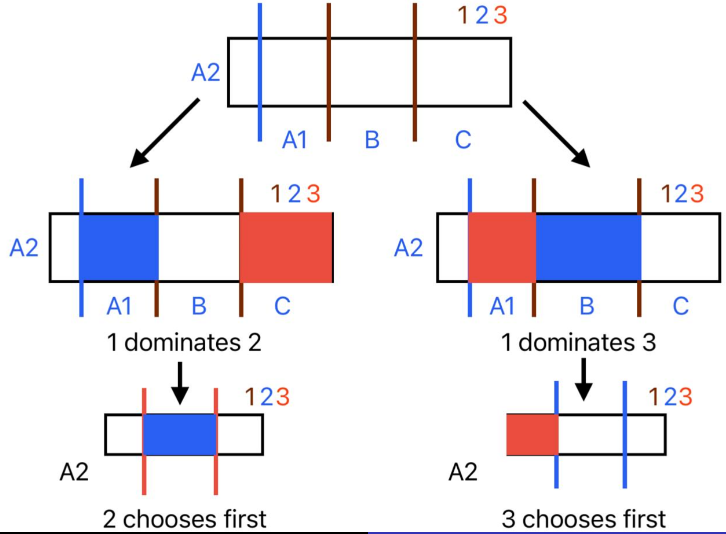

Outline

Austin's Moving Knife - 2 Player Game

We explore whether it is possible to split a cake into n pieces such that both players value each piece at exactly $1/n$

$n = 2$ **(CUT**$(1/2)$**)**

Let P_1 mark a point a such that:

$$V_1([0, a]) = V_1([a, 1]) = 0.5.$$

Assume: $V_2([0, a]) > 0.5 > V_2([a, 1])$ (WLOG)

We then slowly vary the interval $[x, y]$, where initially $x = 0$ and $y = a$, such that: $V_1([x, y]) = 0.5$ This can be imagined as a discrete process where:

$x \leftarrow x + dx, \quad y \leftarrow y + dy, \quad$ such that $V_1([x, x + dx]) = V_1([y, y + dy])$

$V_2([x, y])$ starts at some value > 0.5 and ends at a value < 0.5 when $x = a$ and $y = 1$. By the Intermediate Value Theorem, there exists a point where: $V_2([x, y]) = 0.5$

Austin's Moving Knife - 2 Player Game

Extending to $n > 2$ (CUT($1/n$))
First, we aim to find a single piece that both players value at $1/n$. P_1 cuts the cake into n pieces of equal value. If no piece has value $1/n$ for P_2, there must exist two adjacent pieces where one has value $> 1/n$ and the other $< 1/n$ for P_2 (due to the normalization property).
Apply the CUT($1/2$) procedure to the union of these two adjacent pieces. This yields a single piece valued at $1/n$ for both players.
Now, recursively apply CUT($1/(n-1)$) to the remainder of the cake to continue dividing it among the players. Envy-freeness can be guaranteed by having an unknown function (or randomized) mapping pieces to players.

Brams-Taylor-Zwicker Protocol

A non-discrete method for allocating a cake among 4 players. The steps are as follows:

Brams-Taylor-Zwicker Protocol

- Players P_1 and P_2 use the CUT(1/4) procedure to divide the cake into 4 pieces, each perceived as having a value of 1/4 according to their respective valuations.
- Similar to the Selfridge–Conway procedure, P_3 arranges the pieces in order of preference: $\{A, B, C, D\}$. P_3 then divides piece A into two subpieces, A_1 and A_2, such that:

$$V_3(A_1) = V_3(B).$$

- The players choose their preferred pieces in the following order: P_4, P_3, P_2, P_1. If P_4 does not choose A_1, then P_3 is required to choose A_1.

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Brams-Taylor-Zwicker Protocol

Brams-Taylor-Zwicker Protocol Continued

- Without loss of generality (*WLOG*), assume the player who chose A_1 is P_4.
- Both P_1 and P_2 dominate P_4. Now, P_3 and P_2 use the CUT(1/4) procedure on A_2 to further divide it into equal-value pieces.

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General Idea

The players who perform the cuts are given last preference, as all pieces are of equal value to them. Since P_1 dominates P_4, we allow P_4 the first choice. Brams-Taylor also proposd a discrete procedure for the n-player case which involves getting players to dominate each other pairwise. This procedure is finite but has an unbounded number of cuts.

Sperner's Lemma: Approximations for Envy Free Allocation

Introduction:

- Sperner's Lemma is a simple combinatorial result with powerful applications.
- It applies to a triangle T that is triangulated into smaller elementary triangles.
- Vertices of the triangle T are labeled with the set $\{1, 2, 3\}$.

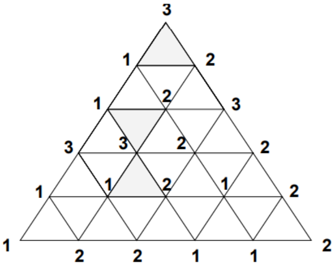
Sperner Labelling

Definition:

- A **Sperner labeling** satisfies the following conditions:
  1. Vertices of the large triangle T have distinct labels $\{1, 2, 3\}$.
  2. Vertices along an edge (i, j) are labeled either i or j.

Illustration:

- Below is an example of a Sperner labeling



Sperner's Lemma: Statement

Statement:

- If the triangulation of T has a Sperner labeling, then there exists at least one triangle labeled $(1, 2, 3)$.

Proof of Sperner's Lemma

Proof Sketch:

- Consider the edges labeled $(1, 2)$ as **doors**.
- Observe that the base of the triangle T has an odd number of such doors.
- A triangle can have:
  - 0 doors,
  - 1 door (if labeled $(1, 2, 3)$), or
  - 2 doors.

Proof of Sperner's Lemma Contd.

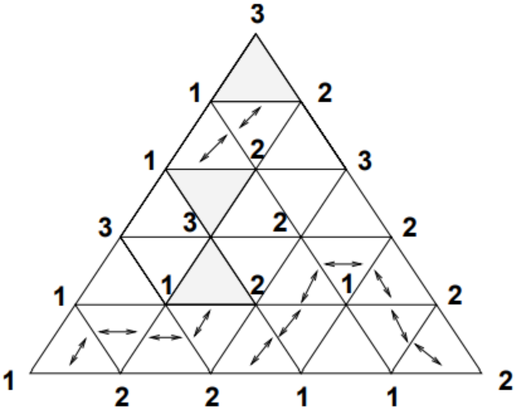
Path Argument:

- Imagine a path starting at the base, entering through a door, and continuing through other doors (without backtracking).
- The path terminates when it:
  - Exits T, or
  - Ends at a triangle labeled $(1, 2, 3)$ (a Sperner triangle).
- For every path that exits T, it must both **enter** and **exit** through doors at the base.
- Since there are an odd number of doors, there must be an odd number of paths terminating at a Sperner triangle.

Illustration of the Proof

Visual Representation:

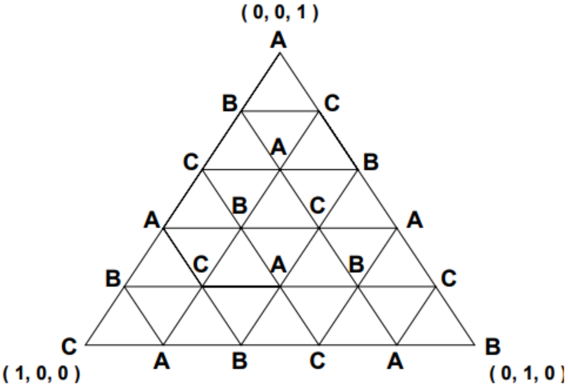
- Below is a depiction of a triangulated triangle T with doors and paths:



Sperner triangulation for resource allocation

Visual Representation:

- Assigning players to vertices

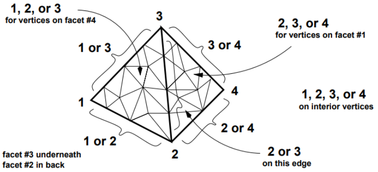


Generalizations to Higher Dimensions

n-Simplex:

- A **0-simplex** is a point.
- A **1-simplex** is a line segment.
- A **2-simplex** is a triangle.
- A **3-simplex** is a tetrahedron.
- In general, an n-simplex is the convex hull of $n+1$ affinely independent points in $\mathbb{R}^m$, where $m \geq n$.

Illustration:

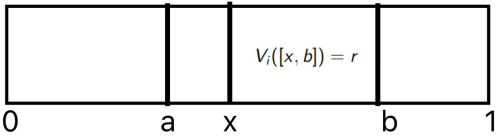


Outline

Major Open Problem: Bounded Protocol for $n > 3$?

- "Question 1: Is there a bounded envy-free protocol for n = 4 or n ¿ 4?" — Brams and Taylor [1995]
- "However, no discrete procedure with a bounded number cuts (however large) is known for four players, and such schemes probably don't exist" — Ian Stewart [2006]
- "one of the most important open problems in the field." — Saberi and Wang [2009]
- "Since the 1940s, the computation of envy-free cake divisions has baffled many great minds across multiple disciplines. Settling this problem once and for all is an important challenge for theoretical computer science." — Procaccia [2013]

Robertson and Webb Model



1. *CUT*$_i$: Given a piece $[a, b]$ and a value $r \in \mathbb{R}^+$, the agent returns a point $x \in [a, b]$ such that:

$$V_i([x, b]) = r$$

2. *EVAL*$_i$: Given an interval $[a, b]$, the agent returns the value of the interval:

$$V_i([a, b])$$

Bounded number of Robertson & Webb queries implies bounded number of cuts

Aziz-Simon Protocol for 3 agents

Let P_1, P_2 and P_3 be the 3 agents

Aziz-Simon 3 agent Envy-free Protocol

- P_3 first divides the cake into 3 equal pieces (A, B, C) according to his value function
- If P_1 and P_2 both want the same piece(WLOG say A:$= [0, a]$), we query $CUT_i(A, \max(V_i(B), V_i(C))$ to both players. Let them return the value x_1 and x_2(Assume $x_2 > x_1$ WLOG)
- The cut portion $[x_1, a]$ is given to P_2, and the trimmed portion $[0, x_1]$(say β) is set aside as residue. P_1 gets his second most preferred piece and P_3 gets a complete piece

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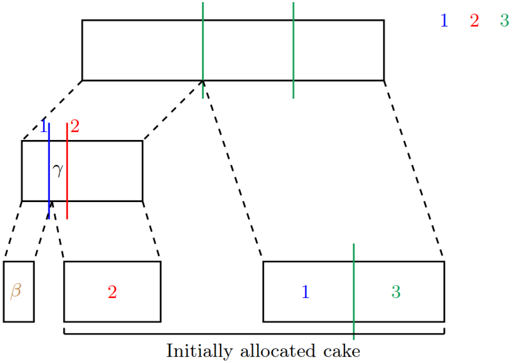
Now this is a partial envy-free allocation

Aziz-Simon Protocol for 3 agents

- P_2 is satisfied with his piece since it has the highest value among all three allocated pieces in his valuation i.e. P_2 gets his second most preferred piece plus a bonus (γ)
- P_1 is envy-free because P_2's piece has the same value as their own
- P_3 divided the pieces and gets a complete piece, so he is not envious



Aziz-Simon Protocol for 3 agents - Idea of Permutation

At this stage, P_3 dominates P_2. If we can ensure that P_3 also dominates P_1, the problem reduces to a two-player game, and we can apply the cut-and-choose procedure to the residue.

Aziz-Simon 3 agent Envy-free Protocol Continued

- Again, let P_3 cut β into 3 pieces, if P_1 and P_2 choose different pieces, we are done, but if they choose the same piece and now P_1 trims more, he gets that trimmed piece plus a bonus, and now P_3 dominates P_1 and we are done.
- Otherwise, if again P_2 trims more and P_2 gets the trimmed piece with bonus(say $beta_1$). If $\beta_1 > \gamma$, P_1 and P_2 exchange the pieces allocated in the first round, otherwise they exchange the pieces allocated in the second round. P_2 still perceives an excess of $|\beta_1 - \gamma|$ over P_1 and P_1 think he and P_2 got the same valued pieces. But now P_3 dominates both P_1 and P_2, and we are done!

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- Otherwise, if again P_2 trims more and P_2 gets the trimmed piece with bonus(say $beta_1$). If $\beta_1 > \gamma$, P_1 and P_2 exchange the pieces allocated in the first round, otherwise they exchange the pieces allocated in the second round. P_2 still perceives an excess of $|\beta_1 - \gamma|$ over P_1 and P_1 think he and P_2 got the same valued pieces. But now P_3 dominates both P_1 and P_2, and we are done!

Key Ideas

- The cutter is always happy to go last as long as he gets a complete piece
- The person to choose first has an irrevocable advantage
- Domination: A player is said to dominate another player if even the whole of residue is allocated to the other player he does not envy him. If we get a player to dominate every other player we can ignore him from the rest of the protocol!
- Permutation: A clever way of reallocating pieces among players $\in N/\{i\}$ to make agent i dominate them

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A Useful Lemma for 4 agents case!

If one player dominates all others we can reduce the problem to 3 players. However, we can do better

Double Domination Lemma

Given a partial envy-free allocation and unallocated cake such that each agent dominate two other agents we can allocate the residue in an an envy-free manner

Proof

We represent the relationships between players as a directed graph, where an outgoing edge (u, v) indicates that u dominates v.

- If a player is dominated by 3 others, the residue can be allocated entirely to this player.
- If 2 players dominate the same pair of players, the dominated pair can use the cut-and-choose procedure to divide the residue.

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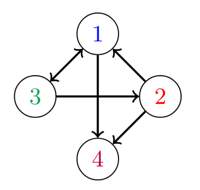
Double Domination Lemma

Proof continued

- The only graph that does not satisfy the above two conditions is represented as follows (subject to permutations of nodes):



- Observe the following:
  - Players 2 and 3 dominate player 1.
  - Player 3 dominates player 2.

Player 4 can divide the residue into 4 equal pieces. The players then choose their preferred pieces in the order $1, 2, 3, 4$, ensuring envy-freeness

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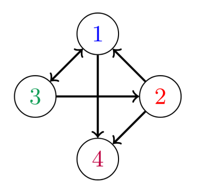

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Core Protocol (i)

Description:

- Agent i is designated as the cutter and divides the cake into four equal pieces.
- The output results in a **partial envy-free allocation**:
  - The cutter and at least one other agent receive a whole piece.
- The piece whose residue has the maximum value according to V_i is termed the **significant piece**.
- The unallocated cake consists of the residue of at most two pieces.

Lemma: Single Domination

Statement: The cutter will dominate the player who receives the significant piece.

Setup:

- Let the residue be $\beta_1 \cup \beta_2$, where β_1 is the residue of the significant piece.

Result:

- If we run the core protocol (4) on the residue:

$$\text{Value of residue} \leq \frac{1}{2}\big(V_4(\beta_1) + V_4(\beta_2)\big) \leq V_4(\beta_1).$$

- This implies that P_4 dominates P_1.

Bounded Protocol for 4 Agents

Setup:

- If player i does not dominate 2 other players:
  1. Run the **Core Protocol (i)** 4 times.
  2. After every 2 iterations, player i dominates the player with the significant piece.

Handling Significant Pieces

Case: If the same player receives the significant piece in all 4 calls:

- Let the bonuses in each round be b_1, b_2, b_3, b_4.
- Identify j such that:

$$2b_j \leq \sum_k b_k.$$

- Run the **Permutation Protocol** on the allocations of the jth iteration of the Core Protocol.

Completing the Allocation

Final Steps:

- Repeat the process for all players i.
- Run the **Post-Double Domination Protocol** to complete the allocation.

Discrete-Bounded envy-free protocol for n-agents

- The very next year after this work Haris Aziz and Simon Mackenzie published a discrete bounded envy-free cake-cutting protocol for any number of agents
- The maximum number of queries required by the protocol in the Robertson-Webb Model is $n^{n^{n^{n^{n^{n}}}}}$ or $n \uparrow\uparrow 6$!!!
- Even if the protocol is not run until completion, it can achieve proportionality and envy-freeness for partial cake in $n^3(n^2)^n$ queries

References

1. A Discrete and Bounded Envy-Free Cake Cutting Protocol for Four Agents - Haris Aziz, Simon Mackenzie (2015)
2. A Discrete and Bounded Envy-Free Cake Cutting Protocol for Any Number of Agents - Haris Aziz, Simon Mackenzie (2016)
3. Cake-Cutting Algorithms: Be Fair If You Can. Natick, Massachusetts: A. K. Peters (1998)
4. Rental Harmony: Sperner's lemma in fair division (Francis Edward Su)